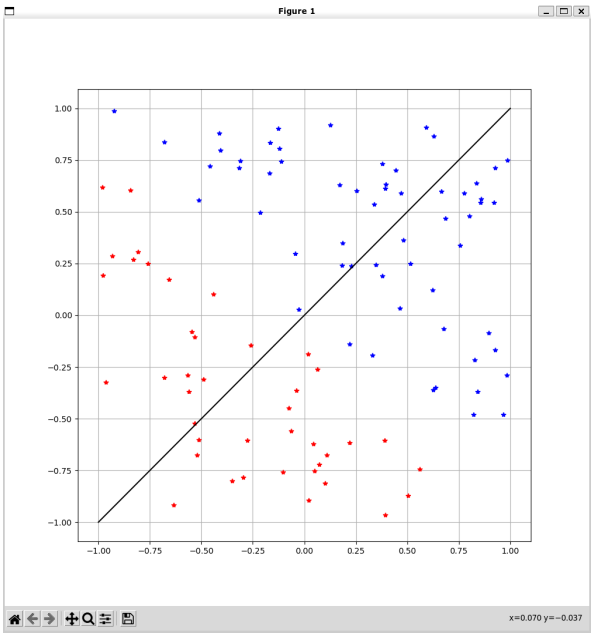
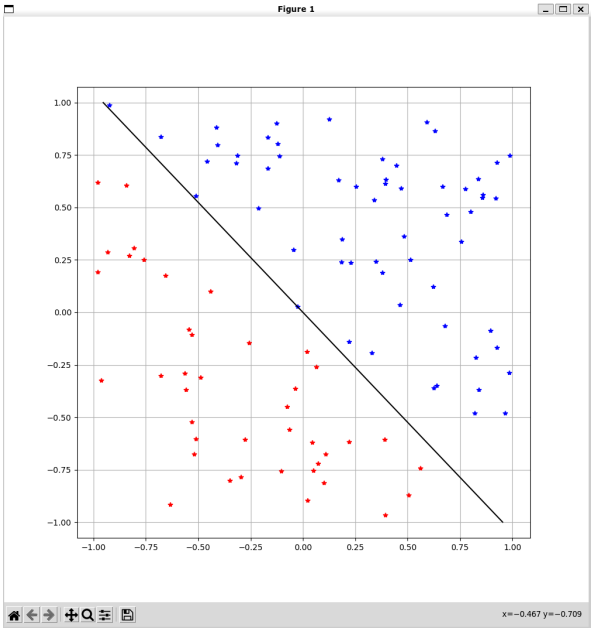
1.
    a. I have not taken any of the following courses, but I have taken recommender systems which has intersections with machine learning
    b. I have taken Introduction to Probability and Statistics (STAT 3021)
    c. I have taken applied linear algebra (MATH 4242) and multivariable calculus (MATH 2263)
2.
    a. X must be an *n* by *k* matrix
    b. If A is not full rank then it does not have an inverse and the problem cannot be solved
3. For vectors *a* and *b* you would use the distance formula on corresponding entries of the vectors: $\sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \ldots}$ The distance between two matrices is the Frobenius distance
4.
    a. no, this method is not accurate because the probability does not directly apply to a sample
    b. (15 points) Another group of researchers developed a new blood test to identify the same disease. The test result is given by a random variable X, with sensitivity and specificity given by 0.7 and 0.8, respectively (that means p(X = 1|D = 1) = 0.7 and p(X = 0|D = 0) = 0.8). If a patient did the blood test and the result is positive, what is the probability that the person has the disease? Hint: you might want to use the Bayes Rule: p(b|a) = p(a|b)p(b
    c. Given that the patient got a positive result, P(D | X = 1) = (0.7 * 0.1) /
5.
    a. 

b.


Figure 1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
x=−0.467 y=−0.709